

A short proof that the planes over $\mathbb{Q}(\sqrt{3}, \sqrt{11})$ and $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ are 4-chromatic

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ABSTRACT. For a subfield $F \subset \mathbb{R}$ let $\chi(F^2)$ be the chromatic number of the graph on F^2 joining points at Euclidean distance 1. Fischer proved in 1994 that $\chi(\mathbb{Q}(\sqrt{3}, \sqrt{11})^2) = 4$. His result seems to have been overlooked: Moorhouse (2010), Madore (2015), Cranston–Rabern (2017), Exoo–Ismailescu (2018) and Voronov (2021) all treat the value as unknown. We give a short proof of a criterion that gives the upper bound 4 for this field and for $\mathbb{Q}(\sqrt{2}, \sqrt{3})$; with a 10-vertex graph it gives $\chi(\mathbb{Q}(\sqrt{2}, \sqrt{3})^2) = 4$, the second case that Voronov thought likely, which we have not found in the literature. The criterion is a case of Madore’s reduction argument for quadratic forms, in other coordinates. With $\alpha = x + y/\sqrt{3}$ and $\beta = 2y/\sqrt{3}$ the squared distance becomes $\alpha^2 - \alpha\beta + \beta^2$, a form that is anisotropic modulo any prime above 2 with residue field $\mathbb{F}_2$.

1. Introduction

The Hadwiger–Nelson problem asks for $\chi(\mathbb{R}^2)$. Since de Grey [2] and Exoo–Ismailescu [4] it is known that $\chi(\mathbb{R}^2) \in \{5, 6, 7\}$; see Soifer [17] for the history. For a subfield $F \subset \mathbb{R}$ one has $\chi(F^2) \leq \chi(\mathbb{R}^2)$. For quadratic fields $\mathbb{Q}(\sqrt{N})$, with N squarefree:

- $\chi(\mathbb{Q}^2) = 2$ (Woodall [19]), and $\chi(\mathbb{Q}(\sqrt{N})^2) = 2$ exactly when $N \equiv 1, 2 \pmod{4}$ (Johnson [9]; Fischer [5], Thm 8);
- $\chi(\mathbb{Q}(\sqrt{N})^2) \leq 3$ for $N \equiv 0, 1 \pmod{3}$, and ≤ 4 for $N \equiv 3 \pmod{8}$ (Fischer [5], Thms 9, 10; see also Payne [13]);
- later, Moorhouse [12] and Madore [11] proved further bounds by reduction modulo a prime.

The field $\mathbb{Q}(\sqrt{3}, \sqrt{11})$ is the smallest whose plane contains a Moser spindle. Fischer [7] proved that the plane over $\mathbb{Q}(\sqrt{p}, \sqrt{q})$ has an additive 4-colouring, with values in $\mathbb{Z}/4$, for squarefree coprime $p \equiv 3 \pmod{16}$ and $q \equiv 11 \pmod{16}$ with $pq \equiv 1 \pmod{32}$; for $\mathbb{Q}(\sqrt{3}, \sqrt{11})$ the Moser spindle gives the lower bound, so $\chi(\mathbb{Q}(\sqrt{3}, \sqrt{11})^2) = 4$. This seems to have been overlooked. Moorhouse [12] noted that the exact value was undetermined, Madore [11] proved $4 \leq \chi \leq 5$, Cranston and Rabern [1] asked for the chromatic number, Exoo and Ismailescu [4] asked whether a 5-chromatic unit-distance graph embeds in $\mathbb{Q}[\sqrt{3}, \sqrt{11}]^2$, and in Polymath16 Voronov wrote: “it seems likely that $\chi(\mathbb{Q}(i, \sqrt{3}, \sqrt{11})) = 4$, and the same is true in case (2, 3). But as far as I know, nobody has proved this yet” [16]. Case (2, 3) is the field $\mathbb{Q}(i, \sqrt{2}, \sqrt{3}) = \mathbb{Q}(\zeta_{24})$, which Fischer’s hypotheses exclude. For it Voronov suggested a colouring through a homomorphism $\mathbb{Z}[\zeta_{24}, 1/3] \rightarrow (\mathbb{Z}/4)[\zeta_{24}]$, adding “Perhaps there is a simpler way” [16].

Theorem 1. $\chi(\mathbb{Q}(\sqrt{2}, \sqrt{3})^2) = 4$. The same argument gives a short proof of Fischer’s theorem $\chi(\mathbb{Q}(\sqrt{3}, \sqrt{11})^2) = 4$.

Both upper bounds are instances of the following criterion.

Theorem 2. Let $L \subset \mathbb{R}$ be a number field containing $\sqrt{3}$, and suppose some prime P of L above 2 has residue field $O_L/P \cong \mathbb{F}_2$. Then $\chi(L^2) \leq 4$.

Theorem 2 is the case $Q = \alpha^2 - \alpha\beta + \beta^2$, with residue field $\mathbb{F}_2$, of Madore’s ¶6.6 [11], which extends his Proposition 3.2 to any quadratic form; we include the short proof for completeness (compare Moorhouse [12], Lemma 4.2 and Theorem 7.1). In the coordinates (x, y) the argument fails at 2: the form $x^2 + y^2 \equiv (x + y)^2$ is isotropic modulo 2, and unit vectors such as $(1/2, \sqrt{3}/2)$ are not even integral at 2. Since $\sqrt{3} \in L$, we have $L(i) = L(\omega)$ with $\omega = e^{2\pi i/3}$, and the hypothesis on P implies that P is inert in $L(i)$.

2. The identity and the lemma

Put $Q(\alpha, \beta) = \alpha^2 - \alpha\beta + \beta^2$. For $x, y \in L$ set

$$\alpha = x + y/\sqrt{3}, \quad \beta = 2y/\sqrt{3}, \quad \text{so that} \quad x^2 + y^2 = Q(\alpha, \beta).$$

The identity is a direct expansion. Equivalently, $x + iy = \alpha + \beta\omega$, and Q is the norm form of $\mathbb{Q}(\omega)$. The map $(x, y) \mapsto (\alpha, \beta)$ is L -linear and bijective.

Lemma. Let v be the normalised valuation of L at a prime P above 2 with residue field $\mathbb{F}_2$. Then for all $(\alpha, \beta) \neq (0, 0)$ in L^2 , $v(Q(\alpha, \beta)) = 2 \min(v(\alpha), v(\beta))$.

Proof. Let $k = \min(v(\alpha), v(\beta))$ and let π be a uniformiser. Write $\alpha = \pi^k \alpha'$, $\beta = \pi^k \beta'$, with α', β' integral at P and not both in P . Then $Q(\alpha, \beta) = \pi^{2k} Q(\alpha', \beta')$. Modulo P , $Q(\alpha', \beta')$ reduces to $s^2 + st + t^2$ with $(s, t) \in \mathbb{F}_2^2 \setminus \{0\}$, since signs do not matter in characteristic 2. This value is 1 for every such (s, t) , because $X^2 + X + 1$ has no root in $\mathbb{F}_2$. Hence $v(Q(\alpha', \beta')) = 0$. ■

3. Proof of Theorem 2

Let $O_P = \{c \in L : v(c) \geq 0\}$ and let $\rho : O_P \rightarrow \mathbb{F}_2$ be reduction modulo P . Fix a set of representatives of the cosets L/O_P , and write $r(c)$ for the representative of $c + O_P$. Since L is countable, no choice principle is needed. Colour $(x, y) \in L^2$ by

$$c(x, y) = (\rho(\alpha - r(\alpha)), \rho(\beta - r(\beta))) \in \mathbb{F}_2^2,$$

which uses four colours. Suppose (x, y) and (x', y') are at distance 1, and let $\Delta\alpha, \Delta\beta$ be the differences of their α - and β -coordinates.

1. By linearity, $Q(\Delta\alpha, \Delta\beta) = 1$.
2. By the Lemma, $\min(v(\Delta\alpha), v(\Delta\beta)) = 0$. So $\Delta\alpha$ and $\Delta\beta$ lie in O_P and are not both in P .
3. Hence $r(\alpha) = r(\alpha')$ and $r(\beta) = r(\beta')$.
4. So the two colours differ by $(\rho(\Delta\alpha), \rho(\Delta\beta)) \neq (0, 0)$. ■

4. Proof of Theorem 1

Upper bounds. Both fields satisfy the hypotheses of Theorem 2.

- **$\mathbb{Q}(\sqrt{2}, \sqrt{3})$.** The prime 2 ramifies in all three quadratic subfields, so no subfield other than $\mathbb{Q}$ is unramified at 2, the inertia group is the whole Galois group, and $2 = P^4$ with residue field $\mathbb{F}_2$.
- **$\mathbb{Q}(\sqrt{3}, \sqrt{11})$.** The prime 2 ramifies in $\mathbb{Q}(\sqrt{3})$ and $\mathbb{Q}(\sqrt{11})$ ($3, 11 \equiv 3 \pmod{4}$) and splits in $\mathbb{Q}(\sqrt{33})$ ($33 \equiv 1 \pmod{8}$). So $2 = P_1^2 P_2^2$, and both primes have residue field $\mathbb{F}_2$.

Both decompositions were also confirmed with sympy and with PARI/GP (§6).

Lower bounds. For $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, a 4-chromatic graph is already implicit in Voronov, Neopryatnaya and Dergachev [18]: their second series of 5-chromatic graphs is built from the 4-chromatic 10-vertex graph $L_{10,2}$ and the unit vectors ζ_{24} and $(\sqrt{6} + i\sqrt{3})/3$, all with coordinates in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. For completeness, here is a short explicit example. Take the unit vectors

$$u_1 = (1, 0), \quad u_2 = (0, 1), \quad u_3 = (-2/3 - \sqrt{2}/6, -2/3 + \sqrt{2}/6),$$

for which $|u_1 + u_2 + u_3|^2 = 1/3$.

Let $P_0 = O$ and $P_k = \sqrt{3}(u_1 + \dots + u_k)$. Join P_{k-1} and P_k by the unit rhombus whose middle vertices are $(P_{k-1} + P_k)/2 \pm u_k^\perp/2$, where $u_k^\perp$ is u_k turned by 90° . The result has 10 vertices and 16 edges.

1. In any 3-colouring, the two middle vertices of each rhombus take two distinct colours, and its two tips both take the third. So each rhombus forces its tips alike, and P_0 and P_3 share a colour.
2. But $|P_0 P_3| = 1$, a contradiction. ■

For $\mathbb{Q}(\sqrt{3}, \sqrt{11})$ the Moser spindle suffices (Moorhouse [12], Prop. 1.4): the unit rhombus with vertices $O, (1, 0), (1/2, \sqrt{3}/2), (3/2, \sqrt{3}/2)$, together with its image under the rotation about O with $\cos \theta = 5/6$, $\sin \theta = \sqrt{11}/6$. The two far tips are at distance 1, since $|(3/2, \sqrt{3}/2)|^2 = 3$ and $2 \cdot 3 \cdot (1 - 5/6) = 1$.

5. Remarks

1. **Prior work.** Fischer's additive colourings [5, 7] came first. He colours the plane through the additive group generated by the unit vectors, the component of the origin, whose translates are the other components [5, Thm 1]. His 4-colouring of $\mathbb{Q}(\sqrt{N})^2$ for $N \equiv 3 \pmod{8}$ is already a reduction at 2, with values in $\mathbb{Z}/4$ [5, Thm 10], and his 4-colouring of $\mathbb{Q}(\sqrt{p}, \sqrt{q})^2$ is also a homomorphism to $\mathbb{Z}/4$ [7]. He

also described the component of the origin in the plane over $\mathbb{Q}(\sqrt{N_1}, \dots, \sqrt{N_d})$ [6]. Moorhouse [12, Lemma 4.2] and Madore [11, Prop. 3.2] pass to the whole plane by cosets in the same way, of a component or of a local ring. Speyer [14] 4-coloured the Moser ring $R = \mathcal{O}_K[1/3]$, $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$, in 2018 by reduction modulo 2: $R/2R \cong \mathbb{F}_4 \times \mathbb{F}_4$, and either projection to $\mathbb{F}_4$ is a 4-colouring. On R the colouring of §3 is one of these projections, written in the coordinates $\alpha + \beta\omega$. An analysis and computer search by Hubai, reported by Gibbs, found that all 4-colourings of R have period 8 [15], and Dúcz [3] has recently given geometric 4-colourings of the Moser lattice and ring. The contribution of this note is the choice of coordinates, which lets Madore's ¶6.6 work at primes above 2 and reduces the question to the decomposition of 2 in L , and the case $\mathbb{Q}(\sqrt{2}, \sqrt{3})$.

2. **Consistency check.** The criterion cannot apply to fields that contain 5-chromatic graphs, and it does not. In $\mathbb{Q}(\sqrt{3}, \sqrt{11}, \sqrt{247})$, the field of Exoo-Ismăilescu, and in $\mathbb{Q}(\sqrt{3}, \sqrt{5}, \sqrt{11})$, the field of Heule [8], every prime above 2 has residue field containing $\mathbb{F}_4$, where $X^2 + X + 1$ has roots.

6. Verification and provenance

- **Upper bounds.** Besides the proof, the colourings were tested by computer on random unit vectors $z/\bar{z}$ of each field and on unit-distance graphs with up to 3 134 vertices, with no monochromatic edge. The prime decompositions were recomputed with sympy and with PARI/GP.
- **Lower bounds.** Both graphs can be checked by hand. An exhaustive computer search also confirms that neither has a proper 3-colouring.
- **Literature.** We have read [5] in full, the zbMATH entries for [6] and [7], and Johnson's paper [9] through its zbMATH entry and Payne's summary [13]. We have not seen the full texts of [6], [7] and [9], nor Johnson's survey [10] of problems in this area.
- **Code and data.** github.com/decalion89/chromatic-number-of-the-plane, release 1.0.0: the proofs are checked by `tests/test_q311.py` and `tests/test_q23.py`, the decompositions by `scripts/decompositions.gp`.

* These results were obtained with the assistance of an AI system (Claude, by Anthropic), which carried out the computations and the checks and helped to draft this note. No mathematician has checked the note yet. The author is not a professional mathematician and submits it for independent verification.

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